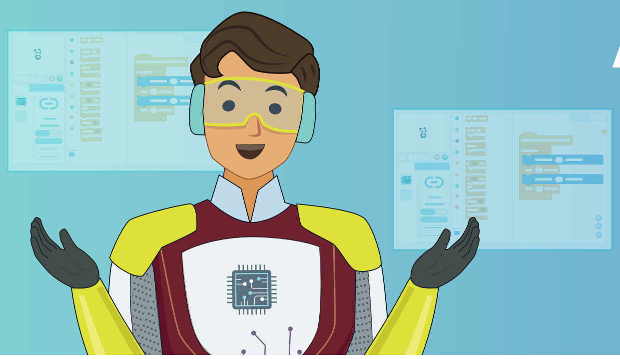
Arduino Programming 13



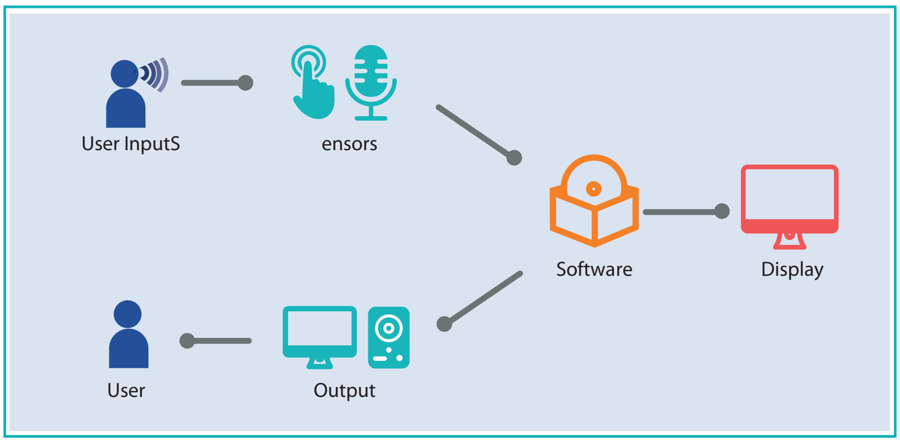
We often hear the phrase, “Data is the new oil”. This is said because every smart device that we use needs data. The process starts with data acquisition, after which devices analyse the data and take actions according to it. Sensors and electronic devices like an Arduino are vital for data acquisition.

13. Arduino Programming

Data acquisition

Data acquisition is a process that enables devices to collect data (numbers, words, images, etc.) and convert it into a digital format. Data needs to be converted into digital form to be analysed by a software and stored in a computer.

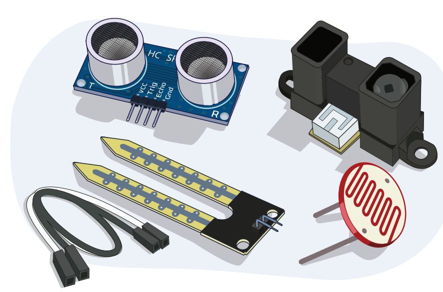
Sensors are used to collect data.



Sensors

A sensor is an electronic device that detects and measures changes in the nearby environment and sends that data to the operating system or processor. It converts physical inputs into electronic signals that are in a human-readable format.

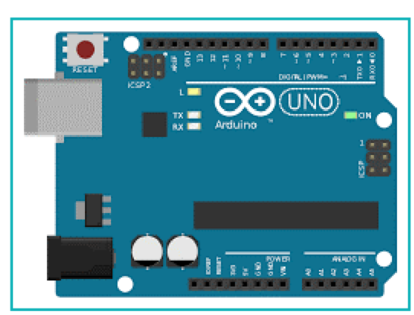
Sensors read data from the environment and relay it to a computer. But a computer doesn't connect directly to the sensor, it needs a mediator. To process and transfer data we need an additional device. An Arduino board is a device that communicates between sensors and a computer.



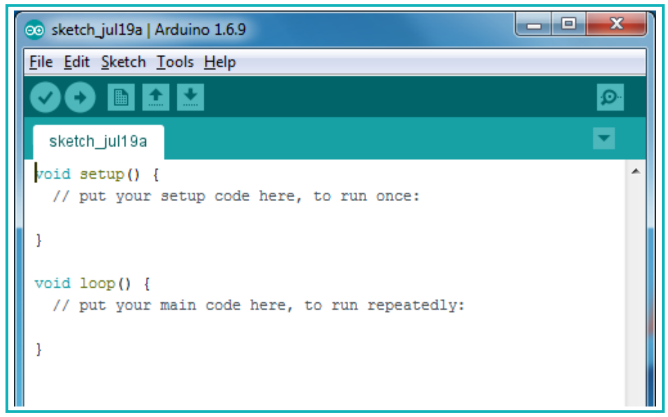
Arduino

Arduino is easy to code, making it perfect for beginners. We use it to create stand-alone circuits.

In the earlier lesson, we made prototype circuits on a breadboard. These circuits can be programmed to perform tasks with a programmable processing board. Arduino is one such board.



Arduino Board (Hardware)



Arduino IDE (Software)

Arduino is a tool to develop interactive devices. It takes inputs from different sensors and sends output in the form of light, motor movement, etc. Arduino is an open-source platform based on easy-to-use hardware and software.

Here, the hardware is a processing board which is programmed using software. On the Arduino platform, the hardware is an Arduino Board and the software is Arduino IDE. Arduino IDE is a text-based programming software which only works with Arduino.

Arduino IDE or block-based programming software is used to read data from an Arduino board.

Key features of Arduino:

- Easy to upload a new code to the same board with a USB cable.
- The Arduino IDE uses a simplified version of C++ (programming language), which makes it easy to learn. We can also program it using block-based visual coding software.

Different Arduino boards are available depending on their processors. Arduino UNO board is one of the most popular.

Arduino Uno is a good board to start with for electronics and programming. A **USB cable** is used to provide power supply and upload code to the board.

Activity

Create a blinking LED project using an Arduino UNO board

To understand the Arduino Uno board, we will use it to create a blinking LED. A blinking LED is used to interpret data from sensors. We have already made a circuit on a breadboard, now let's build a stand-alone circuit to make an LED blink (LED ON/OFF).

This activity requires the following components:

a. Arduino UNO -1

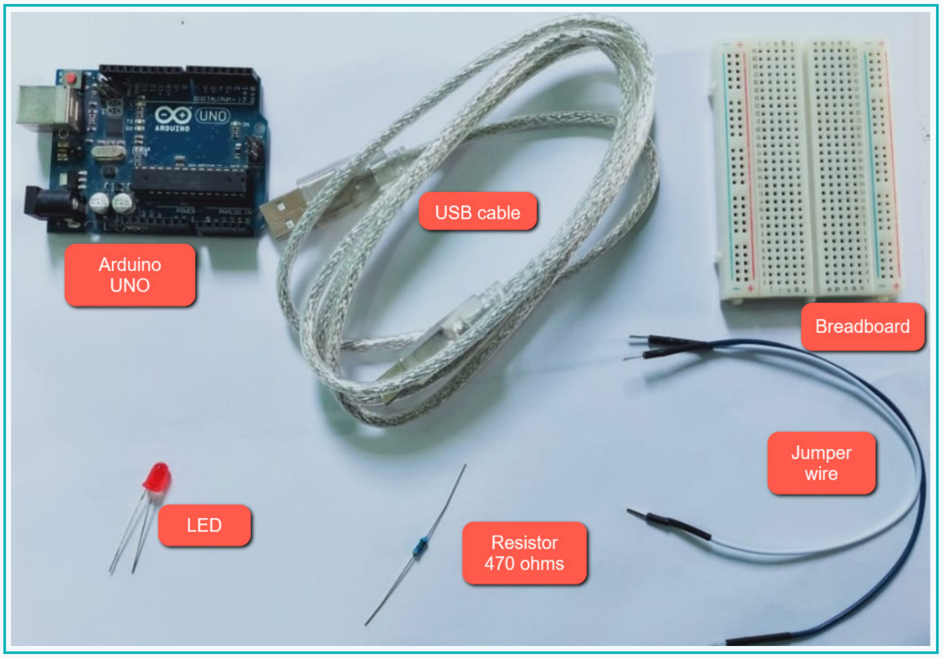
b. USB cable – 1

c. Breadboard - 1

d. LED - 1

e. Resistor – 470 ohms

f. Jumper wires – 1 pair

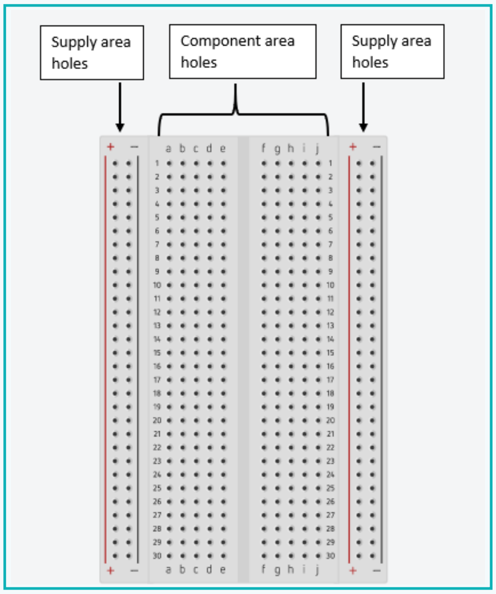


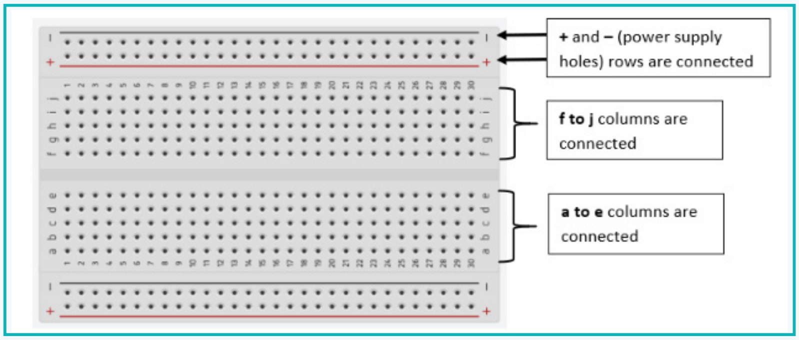
Connections:

We will use a breadboard to connect components and create a circuit.

Activity

Breadboard: A breadboard has holes to place components on it, which are used to create circuits. Breadboards have four sections: two for power supply and two for component connections.

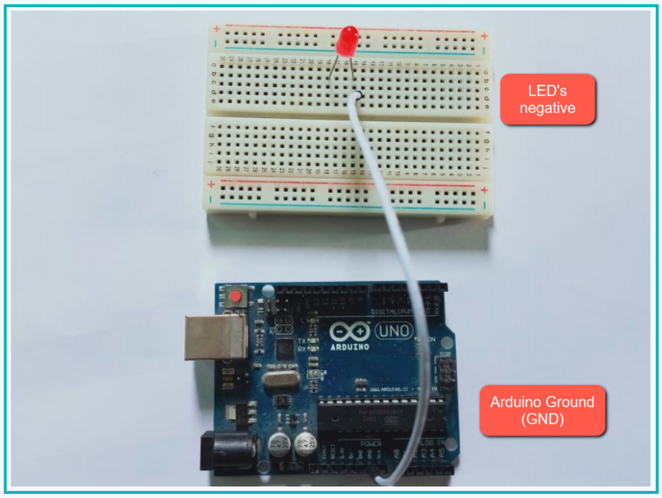




1 Connect an LED to the breadboard

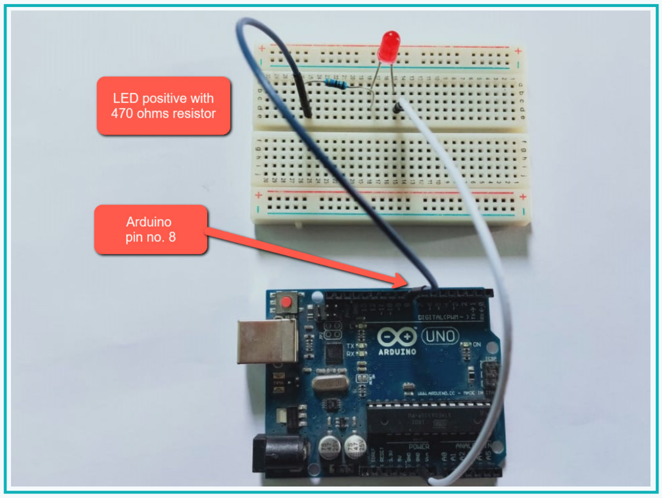
An LED reads digital data in the form of high (ON) and low (OFF).

a. Connect the LED's negative terminal to the GND pin on Arduino (In an LED, the long leg is positive and the short one is negative).



b. Connect the LED's positive pin to pin number 8 (or any other digital pin) using a 470ohms resistor.

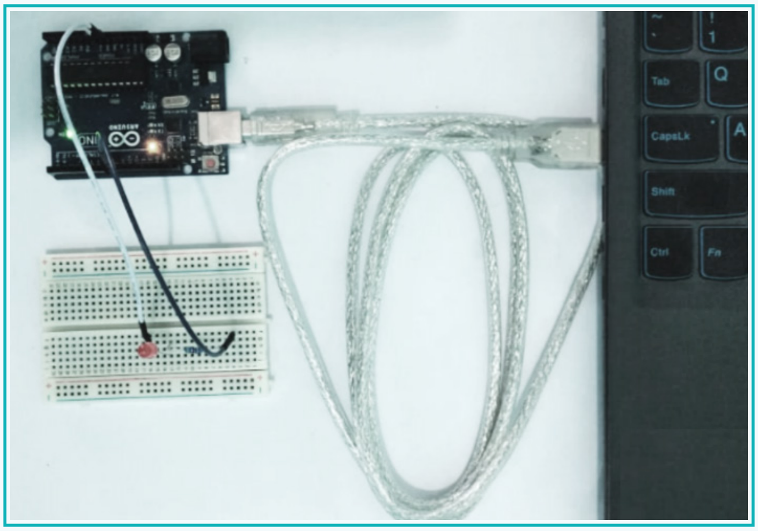
Activity



An Arduino UNO board has 14 digital input-output pins, numbered from 0 to 13. These pins are used to read digital input and provide an output, like an LED.

2 Use a USB cable to connect the two devices

a. Connect a USB cable to the Arduino board's power jack and to the computer's USB port.



b. When the power LED indicator turns ON, on the Arduino board, the LED will turn ON.

Exercise

Tick mark ☑ the correct answer

1. Which of the following is used to upload code to an Arduino board?

   a. LED ☐ **b.** Breadboard ☐
   c. USB cable ☐ **d.** Resistor ☐

2. Which of the following is a part of a breadboard?

   a. Supply Area ☐ **b.** LED indicator ☐
   c. USB Cable ☐ **d.** Jumper wire ☐

3. What is data acquisition?

   a. A process ☐ **b.** A tool ☐
   c. A software ☐ **d.** A device ☐

Answer in one line

4. What is a sensor?

   ____

   ____

Activity

3 Make the LED blink

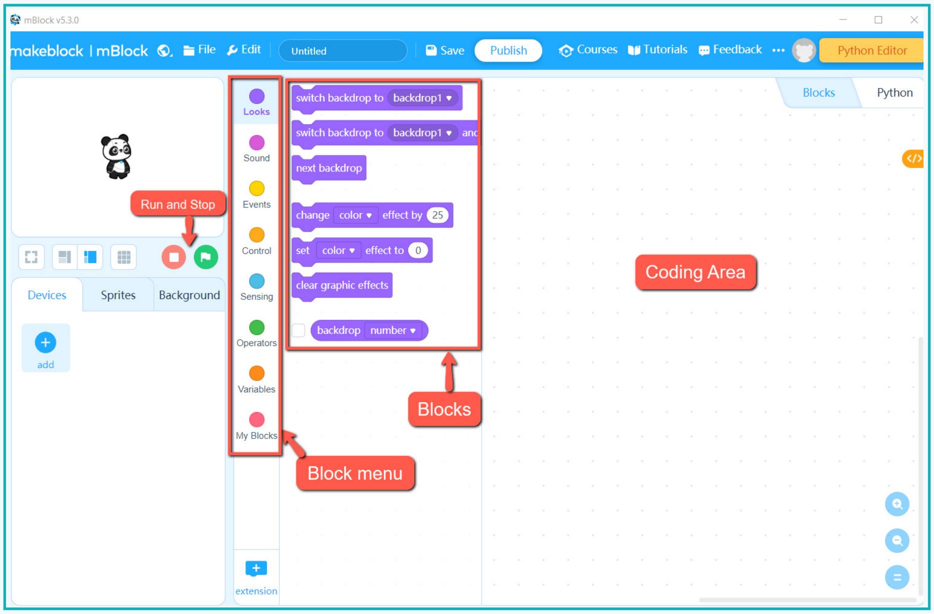
Let's try to make the LED blink once (on/off) after 2 secs, and then blink continuously.

We need to give instructions to the Arduino board. We will use a block-based coding software, **mBlock**, to upload instructions (code) to the Arduino board.

mBlock: Makeblock is a coding platform for beginners, which supports coding for hardware. mblock has the option to select a coding language. We will use block-based programming.

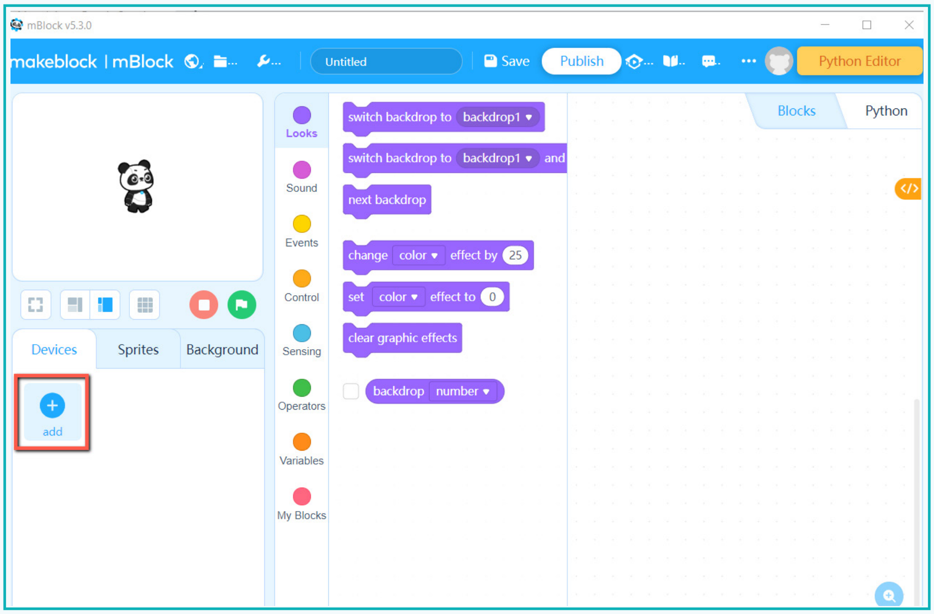
Activity

a. Open mBlock on the computer.



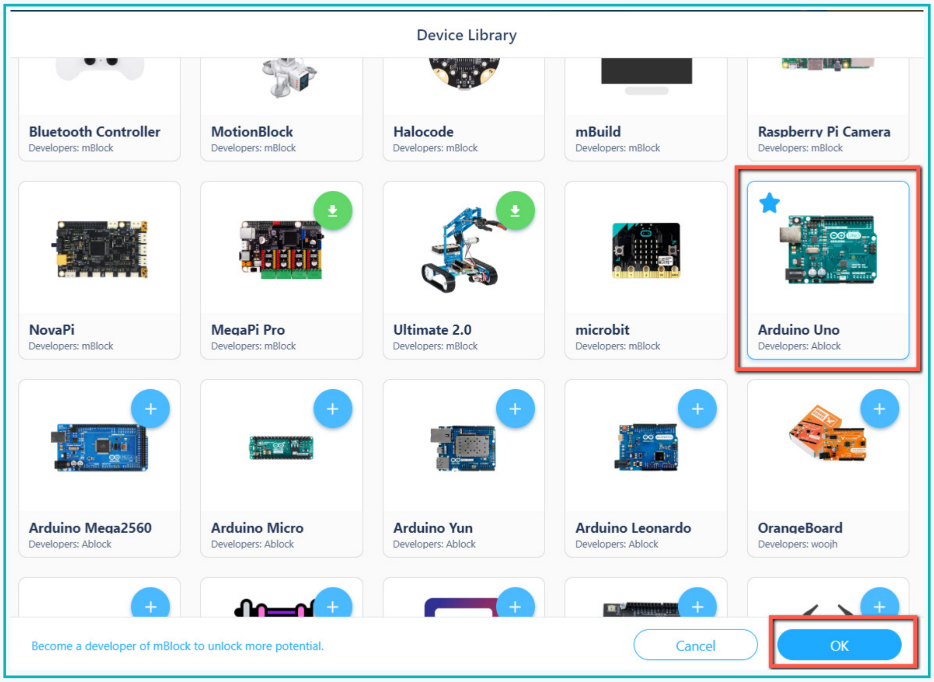
mBlock's first screen

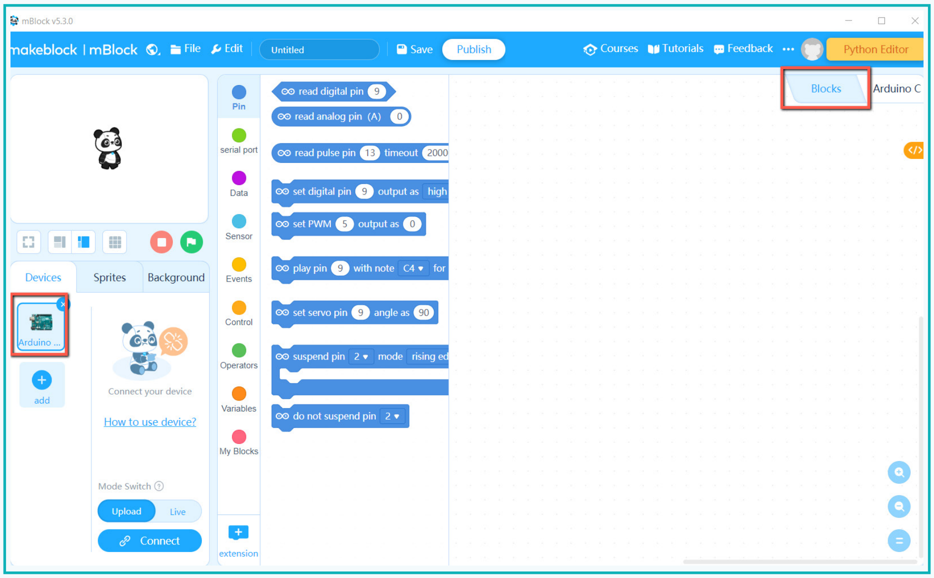
b. Add an Arduino board to use it with mBlock.



Activity

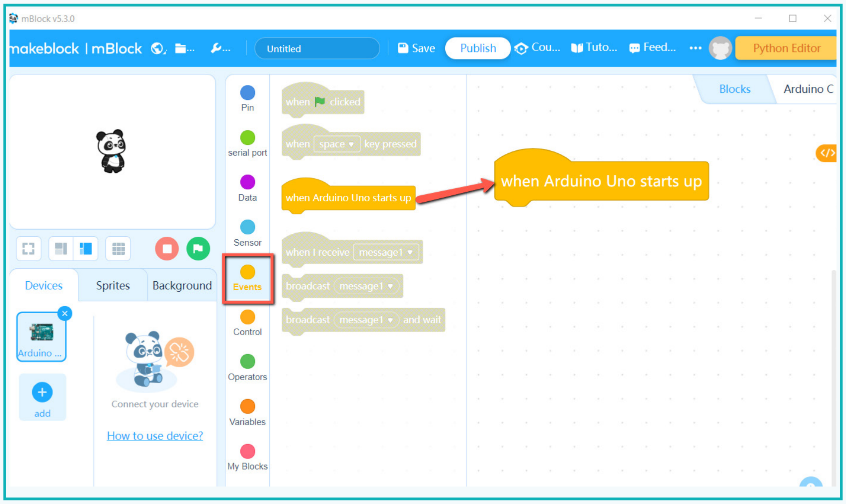
c. Select Arduino UNO.





Activity

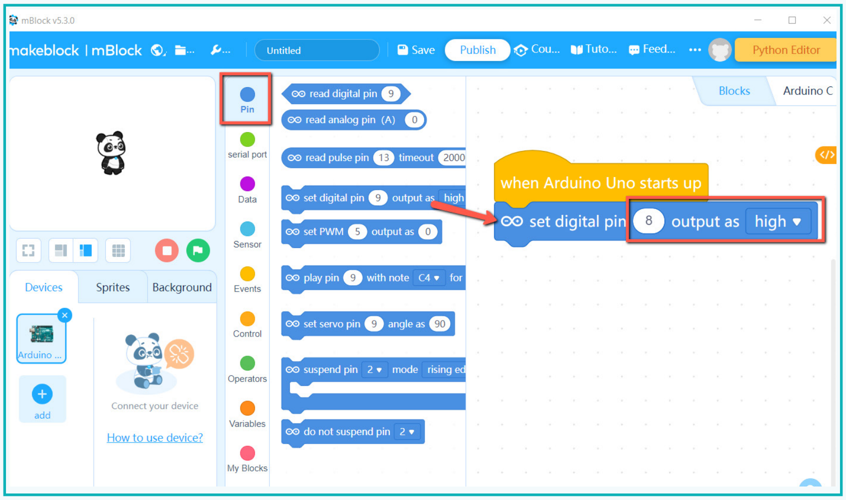
d. From the Event block, add block **"when Arduino UNO starts up"**. This will run when the Arduino board connects to the computer.



e. Since the LED's positive pin is connected on pin number 8 of the Arduino board, we need to check the data on pin number 8. Add **"set digital pin 8 output as high"** (high means LED ON) from pin block.

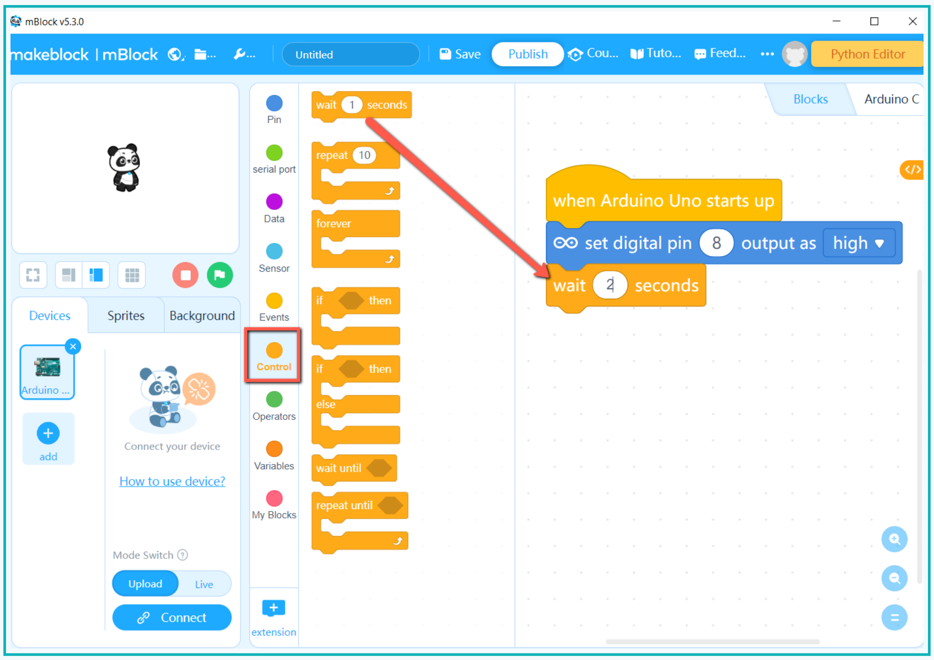
Note:
If an LED is connected on any other pin, then write the corresponding number instead of 8.

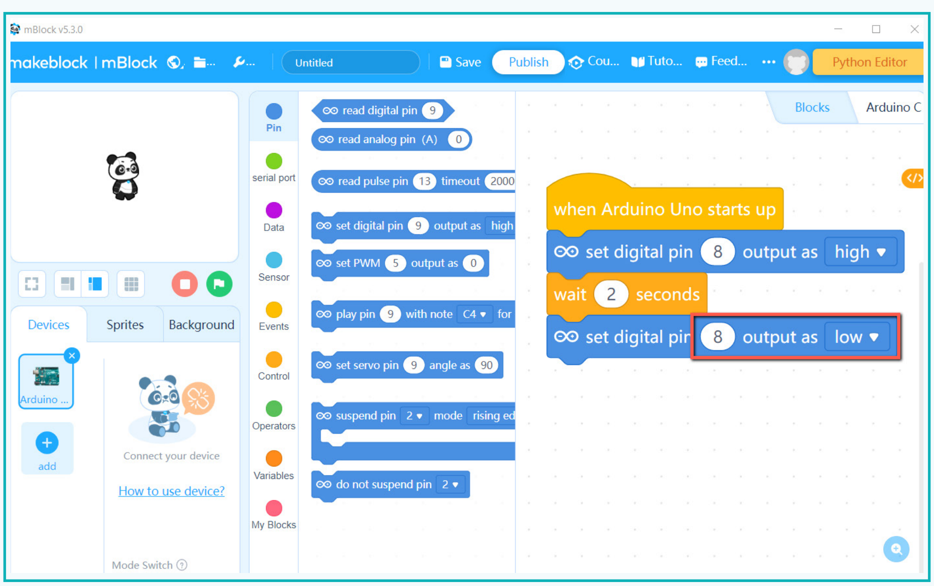
Digital data: Information is stored on a computer in the form of 0's and 1's. 0 means low and 1 means high.



Activity

f. To change the function from on to off, add block **"wait for 2secs"** and block **"set digital pin 8 output as low"** (low means LED OFF) to turn the LED OFF.

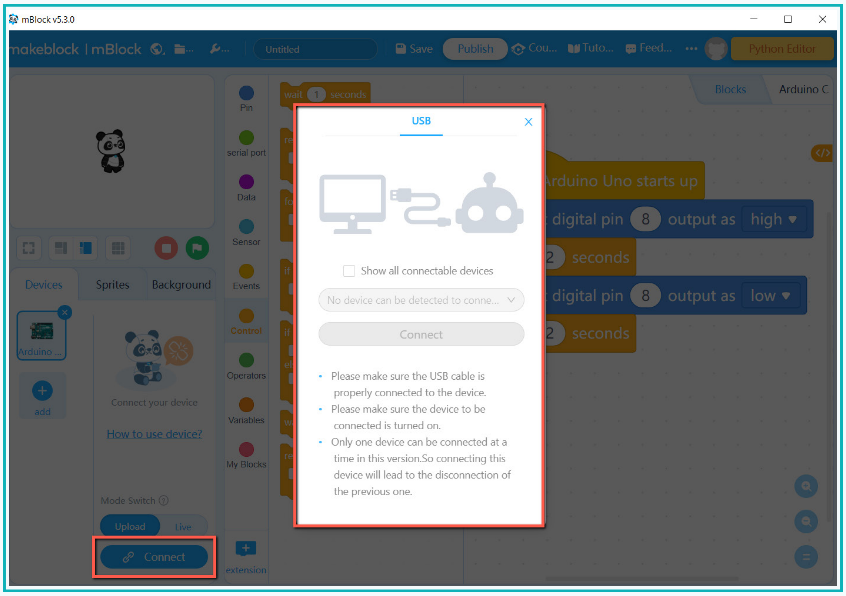




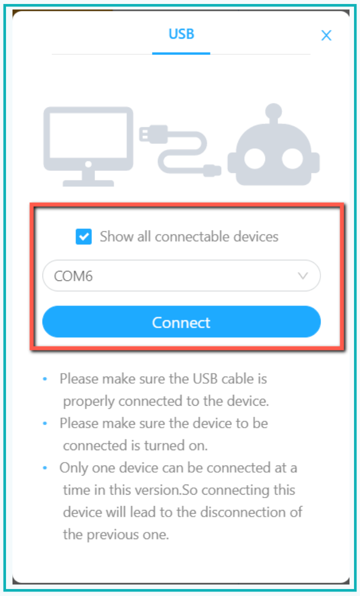
g. To check the output on the Arduino board, connect it to the computer with the USB cable.

Activity

h. Select the box to view all connected devices.



i. Select COM port from the dropdown list.

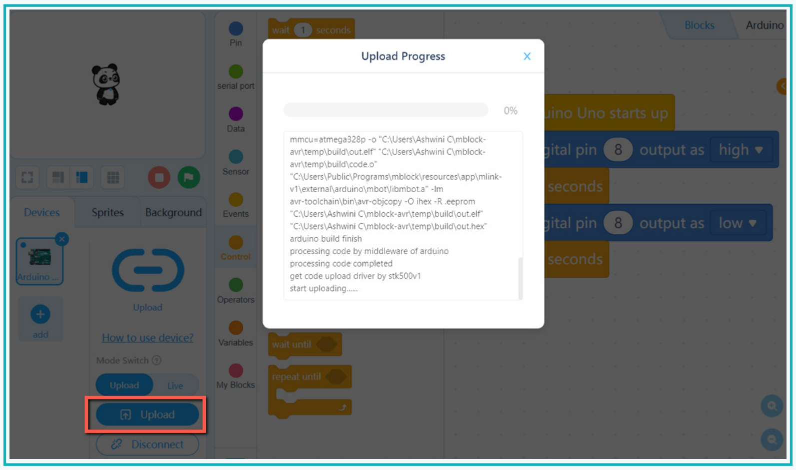


j. Click on connect and wait for a few seconds for the 'connected' message to show on screen.

k. The device is ready to upload the code and check the output.

Activity

l. Click on **upload** and wait for the code to upload to the Arduino board.



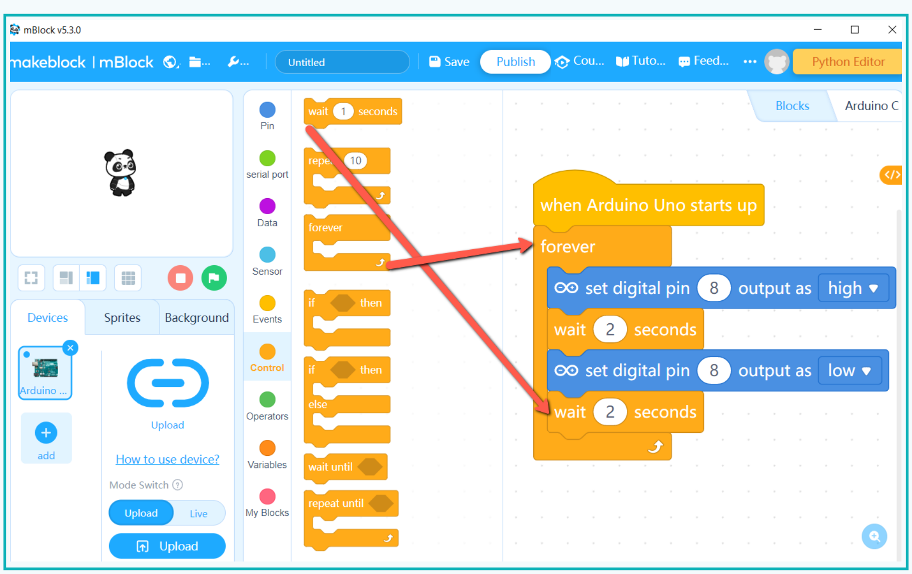
Pop up for upload progress

Observe that the board LED is turned ON and it will go OFF after 2 seconds.

> **Note:**
> If 2 seconds is a short period, increase the number of seconds to see the difference. Upload the code again.

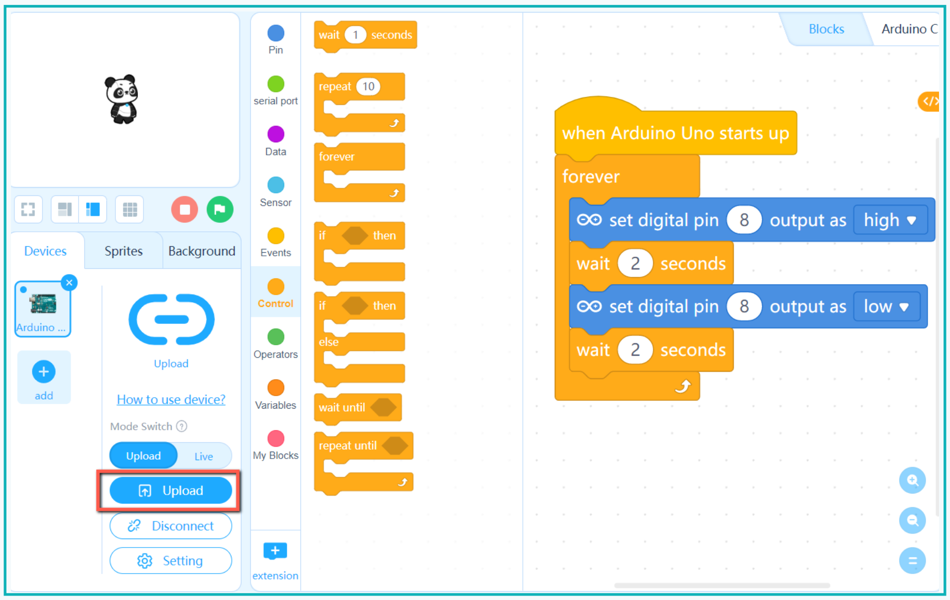
m. To make the LED blink continuously, add forever from the control block. Forever block is used to repeat the process in a loop.

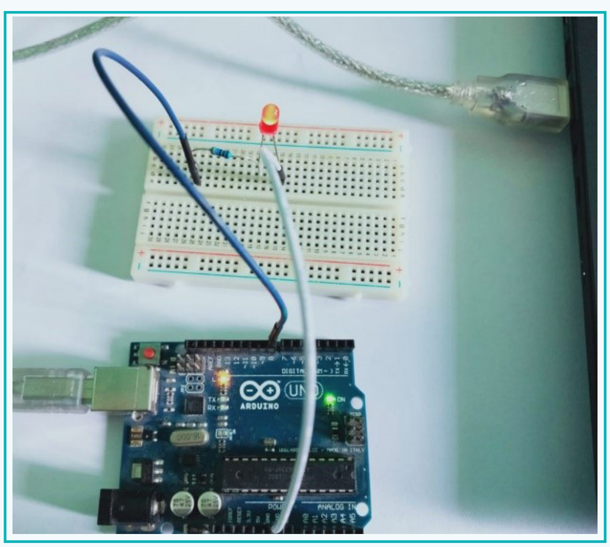
n. To watch it turn ON/OFF, add "wait for 2 seconds".



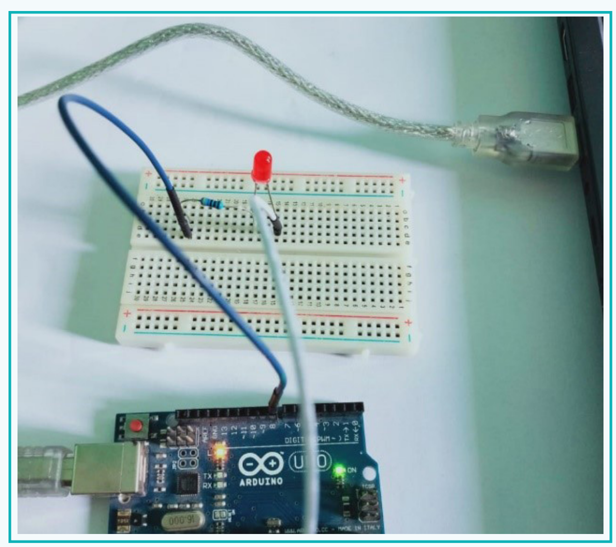
Activity

o. Upload the code to the Arduino board and view the output.





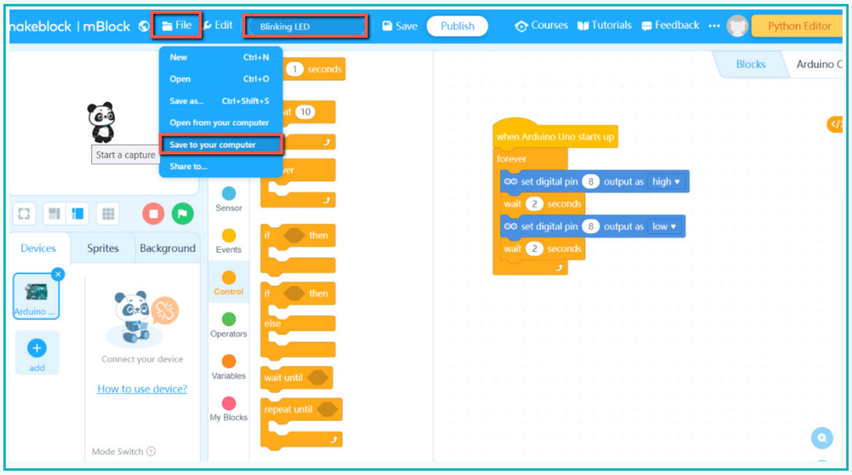
LED ON



LED OFF

Activity

p. To save the code, click on **file** and **save to your computer.** Name it “Blinking LED”.



q. Make a folder with your name and save it.

In this lesson, we learnt about data acquisition and its importance in the working of smart devices. We saw how an Arduino plays a role in analysing data and providing the desired output. We also learnt about the mBlock platform that provides a simple way to program an Arduino.

Exercise

Fill in the blanks

5. An Arduino board works as __________________.

6. An Arduino board can be powered using a __________________ cable.

7. mBlock uses two programming languages which are __________________ and __________________ based.

Exercise

State whether True or False

8. Arduino UNO board connects only via digital sensor.

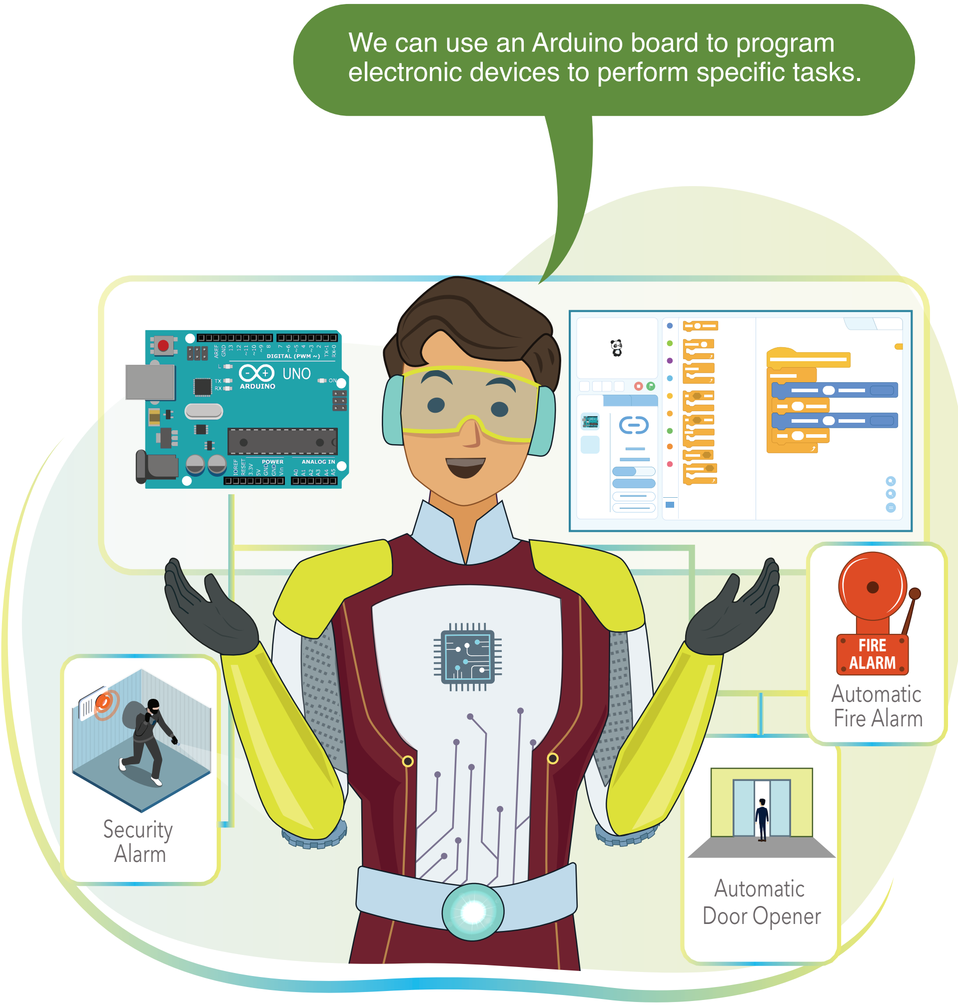
▲ ______________________

9. An Arduino board can follow an instruction without uploading code.

▲ ______________________

10. The power LED indicates if an Arduino board is connected to laptop/PC.

▲ ______________________



Tech Flash

- **Arduino** : Arduino is an open-source platform based on easy-to-use hardware and software.
- **Arduino Uno** : One of the best boards to start with for electronics and programming.
- **USB Cable** : A USB cable is used to power an Arduino board and upload code to it.
- **mBlock** : Makeblock is a coding platform for beginners, which supports coding for hardware. mBlock allows us to use block based and text-based programming.